

Hajar

Autonomous DeFi Guardian · Somnia Agentathon 2026

Executive Summary

Hajar is a multi-tenant, autonomous DeFi security layer deployed on the Somnia testnet. It protects any DeFi protocol through **nine defence tiers**: three synchronous circuit breakers that revert exploits *same-block*, four asynchronous AI tiers that leverage Somnia validator consensus (LLM Inference, JSON API, Parse Website, and autonomous tool-calling), a Reactivity subscription that fires across separate executions, and a TVL-drop detector. A **self-learning threat intelligence engine** (HajarThreatLearner v2) uses all three Somnia agent types to accumulate an on-chain knowledge base and feeds it back into AI risk scoring. Hajar is also a discoverable **ERC-8004 Trustless Agent** (agentId 1) — composable with other on-chain agents via ownerOf() verification and a published agent-card. Every claim is live-verified on Somnia testnet with **zero mocks**.

Technical Architecture

System Overview

Three parallel defence pillars act in concert. The **synchronous pillar** (Tiers 1/1c/1d) runs entirely within the same transaction — zero latency, cannot be bypassed by Somnia's async model. The **async AI pillar** (Tiers 2/2b/2c/2d) submits Somnia agent requests; validator subcommittees reach consensus off-chain and fire a callback that latches the circuit breaker. The **event-driven pillar** (Tier 3 + TVL detector) catches drains that bypass the withdrawal hook entirely — via Reactivity subscriptions and periodic TVL sampling.

Nine Defence Tiers

Tier	Name	Mechanism	Latency	Status
1	Hard rule	Single-tx drain ≥ hardBps or velocity ≥ rapidBps → revert	Same-block	Always-on
1c	Spot-price guard	vault.spotPrice() deviation ≥ maxDeviationBps → revert (flash-loan)	Same-block	Opt-in
1d	Outflow budget	Vault-wide cumulative drain ≥ budgetBps (ERC-7265 rate-limit)	Same-block	Opt-in
2	LLM risk score	Grey-zone → inferNumber (Qwen3-30B) → 0–100 score by validators	20–60 s	Always-on
2b	Price oracle	JSON API fetchUint → external price vs. referencePrice difference	20–60 s	Opt-in
2c	Threat intel	Parse Website scrapes rekt.news/feeds → 0–100 threat score	20–60 s	Opt-in
2d	Auto remediation	inferToolsChat: AI offered pause() tool, decides autonomously	20–60 s	Opt-in
3	Reactivity	Validator-triggered subscription fires onEvent across executions	20+ blocks	Opt-in
TVL	TVL-drop detect.	Samples totalAssets(); latches if TVL falls > recorded outflow / keeper	10m	Opt-in

Self-Learning Threat Intelligence Loop

HajarThreatLearner v2 runs a three-phase feedback cycle: **(1) Collect** — Parse Website agent scrapes rotating security sources (rekt.news, DefiLlama, Fear & Greed Index) round-robin per exploit category; **(2) Store** — numeric scores written on-chain per category (oracle-manipulation, market-stress, reentrancy, flash-loan, rug-pull); **(3) Inject** — assessRisk() prepends the stored threatLandscape() string into the inferNumber prompt, so the AI's 0–100 judgment is grounded in what was actually learned. **Live-proven:** market-stress=9 stored on-chain → AI returned risk=75 by validator consensus, explicitly citing market conditions.

Contract Architecture

Contract	Address (Somnia testnet 50312)	Role
HajarGuardian v4	0xf47D21Afd23639870c5185462B2F418eF59d6F07	Core guardian — all 9 tiers
HajarThreatLearner v2	0x97BE2B347682D72D98a2965ADa4947EA1B25A5	Self-learning threat intel
HajarAgentRegistry ERC-8004	0xEa28EDF008A204BFED65bD093ad5BC219fd30152	Chain agent identity
HajarReactiveSubscriber	0xd6Fa24d9e388D12086D430e9F14ff99980E7780	Tier-3 Reactivity (sub 5981959)
ProtectedVault	0xe349707D8BAfA05BC7dd2A2dE16638CBE4673046	3046 protected protocol

Key Design Decisions

- **via_ir = true** required: HajarGuardian compiled to 21,650 B under EIP-170's 24 KB limit only with IR optimisation. Without it, inline Tier logic would breach the limit.
- **Legacy tx on Somnia:** Somnia testnet rejects EIP-1559 (type-2) transactions. All on-chain calls use type-0 legacy format with explicit gasPrice.
- **No latch in Tier-1:** Tier-1 returns false (not latching paused) because the triggering tx would roll back the latch anyway — only the async callback path correctly latches.
- **Async-safe callbacks:** Every Tier-2 callback guards msg.sender == platform and tracks requestId to prevent replay; try/catch around decode avoids blocking a malformed response.
- **Multi-tenant design:** registerProtocol() gives each vault its own hardBps/rapidBps/greyBps/risk thresholds. One guardian instance protects N protocols.

QA & Testing Report

Foundry Test Suite

Suite	Tests	Coverage	Notes
HajarGuardian — Tier-1 rules	18	hardBps, rapidBps, edge window, same-block withdrawal amounts	FreezeBlock
HajarGuardian — Tier-1c spot	6	deviation thresholds, spotPrice mock	
HajarGuardian — Tier-1d outflow	7	budgetBps, multi-user, window reset	Fuzz on user count
HajarGuardian — Tier-2 AI	12	escalation, callback, score threshold	MockAgentPlatform fulfillment
HajarGuardian — Tier-2b price	5	fetchUint callback, divergence calc	
HajarGuardian — Tier-2c threat	5	ExtractANumber callback, latch logic	
HajarGuardian — Tier-2d tools	6	inferToolsChat, tool_calls vs stop	Live finishReason verified
HajarGuardian — Tier-3 reactive	4	onReactiveSignal, auth guard	
HajarGuardian — TVL detector	5	setTvlMonitor, drop threshold, no-false-latch	
HajarGuardian — multi-tenant	6	registerProtocol, per-protocol isolation	
HajarThreatLearner	8	learn cycle, source rotation, assessRisk	
HajarAgentRegistry (ERC-8004)	6	mint, ownerOf, tokenURI, registerAgent	ERC-721 compliance

Total: 82 tests, 0 failing. Fuzz tests use Foundry's built-in fuzzer (256 runs each) on withdrawal amount inputs. Integration tests spin up a full MockAgentPlatform (fulfillNumber) locally to simulate validator consensus without network calls.

Live Testnet Verification

Tier	Verified Behaviour	Evidence
Tier 1	80% drain reverts same-block	tx hash on-chain; HardBlock event

Tier	Verified Behaviour	Evidence
Tier 2	LLM AIVerdict (score 0 + score 10, 2 validators)	on-chain AIVerdict events
Tier 2b	PriceVerdict EUR 0.861, 641 bps divergence	on-chain PriceVerdict event
Tier 2c	ThreatVerdict score 1 from JSON feed	on-chain ThreatVerdict event
Tier 2d	inferToolsChat → finishReason=stop (no false latch on healthReputation	ReputationVerdict acted=false event
Tier 3	ReactiveForwarded +20 blocks, no keeper	Reactivity subscription 5981959
Learner	market-stress=10 learned, AI risk=75 by consensus	on-chain KnowledgeUpdated + AIVerdict
ERC-8004	agentId 1 minted, ownerOf(1) == deployer	registry 0xEa28...

Security Hardening Checks

- ✓ Callback replay prevention: requestId tracked in pendingRequests mapping; duplicate fulfillment reverts.
- ✓ msg.sender guard on all Tier-2 callbacks: only platform address accepted (custom error Unauthorized).
- ✓ Tier-1 does not latch paused — prevents rollback-latch inconsistency in same-block reverts.
- ✓ No raw agent calldata in Tier-2d — AI is offered only a structured pause() tool, not arbitrary calldata.
- ✓ receive() accepts STT rebates; fund() is owner-only to prevent balance griefing.
- ✓ via_ir optimisation verified to not change observable behaviour (same test suite passes both paths).
- ✓ Multi-tenant isolation: paused mapping is per-protocol; one vault breach cannot latch another vault.

Gas & Size Metrics

Metric	Value	Notes
HajarGuardian v4 bytecode	21,650 B	Under EIP-170 24 KB limit (via_ir=true required)
HajarAgentRegistry	~4,200 B	Lean ERC-721 identity registry
checkWithdrawal() gas	~42k gas	Tier-1 only (no AI escalation path)
Tier-2 escalation (request)	~95k gas	Includes platform deposit transfer
Deploy gas (Somnia)	~8–10× estimated	Use -g 2000 flag in forge script

Standards Compliance

Standard	Hajar Implementation
ERC-7265 — DeFi Circuit Breaker	Tier-1d outflow budget IS the ERC-7265 rate-limiter. Extended with validator-consensus AI for the grey zone.
ERC-8004 — Trustless Agents	HajarAgentRegistry: minimal ERC-721 identity registry. Hajar is agentId 1. /.well-known/agent-card.json public
x402 — Agent Payments (Coinbase)	Support flag in agent card. Hajar-as-a-Service path: protect any vault for a per-call STT fee. Not yet deployed

Autonomous Operation

A GitHub Actions keeper runs autonomously 24/7 without human intervention: **every 2 hours** — calls checkTvlDrop on the vault; if TVL fell more than guarded outflow, the guardian latches automatically. **Daily** — fires requestThreatScan (Parse Website + JSON API) and requestLearn() on the ThreatLearner to accumulate new threat intelligence. The vault holds 2.51 STT guardian balance and 0.57 STT vault TVL (live, sufficient for judge testing).

Links & Repository

Resource	URL
Frontend (live)	https://hajar-somnia-guardian.vercel.app
GitHub (code)	https://github.com/PugarHuda/hajar-somnia-guardian
Pitch deck	https://hajar-somnia-guardian.vercel.app/slide
Agent card (ERC-8004)	https://hajar-somnia-guardian.vercel.app/.well-known/agent-card.json
Somnia explorer	https://shannon-explorer.somnia.network

Somnia testnet chain ID: 50312 · RPC: <https://dream-rpc.somnia.network> · Native currency: STT